Electronic Principles and Devices · Unit_01 · Unit-1 Semiconductor Diode

DRAFT — UE26EC141A · Unit_01 · Unit-1 Semiconductor Diode · Diode Clippers : Series using Ideal diode

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

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Contents

1. 1 · Diode Clippers : Series using Ideal diode
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

Diode Clippers : Series using Ideal diode

1 Diode Clippers : Series using Ideal diode

Study focus: Diode Clippers : Series using Ideal diode. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000070 · EV0000071

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today, we're going to see the clippers.

Key details

- Clippers are those circuits which are made up of electronic components such as diodes, resistors, and DC bias voltages, which are used to remove a part of the waveform without disturbing the other part of the waveform.
- That means to say, in case you have a square waveform, as shown in the figure here.
- If I need only the positive part of the waveform and I don't want the negative part, using the clipper circuit, we can get only the positive part.
- Further, the series clipper can be grouped into unbiased series clipper, biased series clipper, and in parallel, it's been classified into unbiased parallel and biased parallel.
- The series configuration is defined as one where diode is in series with the load.
- Figure 1A, we are using a series unbiased clipper.
- First of all, what is this unbiased?
- Unbiased means there is no DC supply used in the circuit.
- Only we have in this block, we have only diode and the resistor.
- In case if there is any bias voltage here along with the resistor or along with the diode, other than the supply voltage.
- Other than the supply voltage, if we have a voltages here, a DC voltage, then it is biased.
- If the diode is connected in series with the resistor, it is called as a series configuration.
- The response of the series configuration of Figure 1 to the variety of alternating waveform is as shown in the figure B.
- That means to say, variety of the input is first, in the diagram here, which I'm pointing, it is a square waveform.

- The output of the series clipper will be only the positive off cycle, negative has been clipped off.
- Similarly, we have next, in the place of square, we have a triangle waveform.
- That means we need to understand that there's no boundary, where we can also use a sine waveform, the sawtooth waveform, et cetera, as an input signal and based on the circuit, like a clipper, the output variations can be obtained.
- Figure shows here.
- Figure shows a series clipper with a DC bias supply.
- This is what I was referring previously.
- When there is a DC bias here, it is called as a series clipper with a bias or a series clipper with a DC supply.
- This configuration is called as a biased series clipper.
- The addition of a DC supply to the network as shown in the figure can have a pronounced effect on an analysis of a series clipper configuration.
- That means there could be some change in the output of which the DC supply can be in the leg between the supply and the output or in a branch parallel to the output.
- It can be at any place.
- One, it can be between the input and the diode, or diode and the resistor, or resistor and the output.
- It can be anywhere.
- The output response is not as obvious because the DC can aid or work as a source voltage.
- That means the DC source, which is there here, can aid in the sense it can have an addition with the voltage applied here or it can subtract with the voltage applied here.
- That we will understand with the analysis in the next slide.
- The step one as shown here, take careful note of where the output voltage to be defined.

Example or application from the lecture

- Example in the.

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Core explanation · 05:00–09:59

So we need to first note that where we need to calculate the output voltage.

Key details

- Try to develop an overall sense of the response by simply noting that how the input supply affects the conventional current direction through the diode.
- This means to say, check on the input voltage, how it has been applied here, whether it's a positive or a negative.
- And since we are working with an ideal diode here.
- It is an ideal diode.
- Entire discussion will be with the ideal diode.
- Means to say, the diode is on, even if the application of the voltage is zero.

- If it's more than zero.
- Whereas for a non-ideal diodes, it will be 0.7.
- And here, one more thing we also we need to remember, Vi is greater than V.
- Otherwise, the diode will not be on.
- Obviously, this should be there for the diode to be on.
- That means to say, diode is on.
- How do you represent when the diode is on?
- By a close switch.
- That is, VD is equal to zero.
- For ideal diode characteristics, if I look at the ideal diode characteristics, this is what we looked at the previous class.
- This is V and forward bias.
- And diode will start conducting at the zero.
- That's what we have written here, VD is equal to zero.
- Then the current ID,.
- VD is equal to zero, diode is on, but how about the current?
- So note that the substitution of a short circuit equivalent for a diode is as shown in the figure three we discuss now, and the fact that the voltage across the resistor is zero, because the diode current is zero milliamps.
- The result is Vi minus V is equal to zero or Vi is equal to V and is applied a transition voltage.
- This permits drawing a line on the sinusoidal supply voltage as shown in the figure four.
- As shown in the figure four, on state.
- To define the region where the diode is on and off.
- Figure five, the output voltage is defined by, apply the KVL here.
- Apply the KVL in this direction.
- Vi, this is Vi plus Vi.
- Vout is equal to...
- What is Vout voltage?
- Vout is equal to Vi minus V.
- Then for the off-region.
- This is off-region and if you go back, here this is the off-region.
- Off region, diode is in open circuit and ID current is equal to zero milliamps and the output voltage is zero.
- Since there is no current there, output voltage is zero.
- This is shown in the figure.
- It may not be the previous diagram.
- It can also be written in this diagram with the equivalent scales.
- So for on condition, equation one can be used to find the output voltage with the applied voltage as its peak value.
- Output voltage VOUT is plotted as shown in the figure six.
- This is the figure six, how we plot it.

Example or application from the lecture

- Means to say, for example here, this DC voltage, what we apply, and along with the input voltage, whether it make the diode on or off.
- That is given as equation number one here.

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Core explanation · 10:00–14:59

Now let us solve a numerical example one.

Key details

- The question is, determine the output waveform for a sinusoidal input of figure seven.
- This is figure seven.
- I have a sinusoidal signal here.
- VI 20 volts is a peak to peak of one cycle.
- Step two, the positive region of the VI and the DC are both working in the same direction to turn on.
- So the applied total voltage is 10 plus 5 because both are in the same direction.
- We have plus is there, and we have plus is there, it get added.
- The result is that we can safely assume the diode is in on state for entire region of positive cycle voltage.
- Once the supply goes negative, it will have exceeded the.
- DC supply voltage of 5, therefore it would turn off.
- What is the transition model?
- If VI and the DC voltage is greater than the diode cutting voltage, that is VD is equal to zero, then the diode is in on condition.
- Diode is on.
- Therefore, we apply the KVL over there.
- If you apply the KVL over there, VI plus 5 is equal to zero, so VI is equal to minus 5 volts we get.
- Minus 5 volts.
- For a voltage less than 5, the diode is in open circuit and the output is zero, as shown in the sketch of the VOUT.
- This is what we want to prove here.
- Using the figure number nine, we find that for a condition when a diode is on and the diode current is established, the output voltage will be following.
- So the analysis of the clipper circuit with the square wave input is actually easier than the sinusoidal because only two levels are to be considered.
- The transition is important.
- It is clipped off.
- For a transition from 0 to T/2, the figure 11.
- Result, the diode is in short circuit state, that is on.
- If we apply the KVL, we get VOUT is equal to 20 plus 20, 25 volts is the applied voltage.
- Then for VI is equal to -10 during the negative cycle, diode is off.

- That is VOUT is equal to zero.
- Unless we get a positive cycle only, and during the negative cycle, we won't see the output waveform.
- It is clipped off.

Example or application from the lecture

- Then coming to the circuit here, we have a biased clipper here, series biased clipper, where resistor is connected in series with the diode and voltage 5 volts is given, and VI is equal to 20.
- That means to say in the place of sine, if you have a square, only two places if you consider it is more than, whereas here we need to consider for all the places here.
- So this is the solution for example number one, where we see that during the positive cycle, diode is on, during the negative cycle it is off.

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A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Diode Clippers : Series using Ideal diode transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/009_Diode_Clippers_Series_using_Ideal_diode__bd9b40ddcbd2d089.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Diode Clippers : Series using Ideal diode transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/009_Diode_Clippers_Series_using_Ideal_diode__bd9b40ddcbd2d089.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Diode Clippers : Series using Ideal diode transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/009_Diode_Clippers_Series_using_Ideal_diode__bd9b40ddcbd2d089.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9987; only the JSON segmentation supplies the displayed timestamps.
Diode Clippers : Series using Ideal diode transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/009_Diode_Clippers_Series_using_Ideal_diode__bd9b40ddcbd2d089.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
2. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
3. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
4. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00034	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0067. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0068 to s0141. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0142 to s0200. Sidecar variants compare...
SRC00035	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0067. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0068 to s0141. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0142 to s0200. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00036	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0067. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0068 to s0141. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0142 to s0200. Sidecar variants compare...
SRC00037	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0067. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0068 to s0141. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0142 to s0200. Sidecar variants compare...

C Revision and retrieval practice